Matchmaking Engine Audit

Badminton Queue App · 2026-08-12 · verified against source: 54/55 claims confirmed, 0 wrong (4 independent readers, 3 adversarial verifiers)

1 The engine in one paragraph

The matchmaking engine is the app's automatic match suggester. It watches the waiting queue and assembles groups of four into proposed doubles matches. It **never puts anyone on a court itself** — it only fills the “on deck” line with drafts. A separate promotion step moves the front of that line onto a court when a match ends, is cancelled, or the organizer presses **Call Next**. Think of it as a maître d’ who writes seating plans but never walks anyone to a table.

2 When it runs

The engine re-runs after nearly every state change: a player joins, leaves, checks out, or unpauses; a match ends or is cancelled; a draft is published or cleared; auto-matchmaking is toggled on; the draft-cap override changes; or auto-publish is enabled. Each run self-checks first: the toggle must be ON, no other run in flight, at least one court open — and the session's match history must load. **If history can't be read, the engine stops rather than matching blind**: with no history every repeat would look fresh, so it fails closed instead of producing duplicate pairings. All gates fail silently by design; most triggers run the engine in the background after the response is sent.

3 The hierarchy — how a match gets set up

Ordered from strongest rule to weakest. **ABSOLUTE** rules are never waived; **SOFT** rules are preferences that degrade gracefully.

#	Rule	Strength
1	**Permission gates.** Toggle ON, no run in flight, a court open — else nothing happens.	**ABSOLUTE**
2	**History must load.** Failed or over-200-match history stops the run (fail-closed).	**ABSOLUTE**
3	**Draft capacity.** Max 3 outstanding drafts; 5 at 25+ waiting; 6 at 30+. The organizer override can only *lower* it.	**ABSOLUTE** per run
4	**Small-pool patience.** With 4 or fewer waiting and a game in progress, wait up to 8 min for finishers to rejoin. A 2nd draft in one pass needs 8+ waiting. **Call Next bypasses both.**	**SOFT**
5	**Eligibility.** Only *waiting* players — paused and checked-out are invisible, no exceptions. Returning players need 18 min rest, waived entirely when fewer than 4 rested players remain.	**ABSOLUTE** / **SOFT**
6	**Anchor priority.** Everyone is scored in three tiers (table below); the top scorer becomes the *anchor* and the match is built around them.	—
7	**Partner cap of 2.** No pair ever teams up a 3rd time in a session. Anyone at the cap with the anchor is removed from consideration entirely. **The one rule never waived on any path.**	**ABSOLUTE**
8	**Skill window.** All four within ±1 level, widening to ±2 (Red Zone anchors up to ±4). Dropped only when every window fails AND the anchor has waited 15+ min — and even then rule 7 holds.	**ABSOLUTE** per attempt

#	Rule	Strength
9	**Candidate preference.** Prefer fewest games played, least recent overlap with the anchor, longest wait. Rankings, not rules.	**SOFT**
10	**Diversity.** Reject a four where 3+ shared a single recent match (lookback scales 2→7 with pool size). Repair ladder: swap weakest member → wider swap → accept a forced repeat with rotated teams.	**SOFT**
11	**Team split.** Snake draft: most skill-balanced splits first, preferring fresh partnerships and un-repeated opponents (opponent cap of 2 is soft). Lopsided teams only when every balanced split is partner-capped — flagged, and a repair swap tried first.	Balance > freshness
12	**Cross-court pull.** If the waiting pool can only produce a repeat, borrow exactly ONE player still on court (never on a 2-game streak) into a "held" draft, court-eligible only after their game ends + a short rest. Never for the first slot or Red Zone anchors.	Conditional
13	**Draft → publish → court.** Engine output is always a court-less draft. Organizer publishes (or auto-publish skips review). Only the promotion step assigns a court, atomically — two freeing courts can't grab the same match.	**ABSOLUTE** separation

Priority tiers (rule 6 in detail)

Tier	Who qualifies	Score
Hard Cap (top)	Waited 25+ min AND played fewer than 5 games	2000 + 10 per extra minute — the longest waiter always leads
Red Zone	Waited 20+ min	1000 + wait – 8 × games played
Normal	Everyone else	wait – 8 × games played (can go negative)

The 8-points-per-game deduction is the fairness lever: it lets game count drive ordering when wait times are similar, so fresh players get served before someone queueing for their 5th game.

4 How a match reaches a court

```
Engine draft (pending, unpublished, no court)
  → organizer publishes  [or born published in auto-publish mode]
  → published on-deck queue
  → PROMOTION — the only court assigner
     triggered by: match ends · match cancelled · Call Next
     takes the front-most READY published match atomically;
     skips not-yet-ready held drafts; auto-clears matches whose players left
  → in_progress on a court
  → completed (+1 game each, wait clock restarts) or cancelled (no game counted)
```

Player status mirrors this: waiting → drafted (in an unpublished draft) → on-deck → playing → back to waiting. Drafts are invisible to players and the TV until published.

5 Safety rails

- **Race-proof commits.** Match creation runs through a DB transaction with three guards (pre-flight status check, ordered row locks, active-match conflict check). A clash between simultaneous engine runs returns NULL and the engine skips that slot gracefully — double-booking is impossible at the DB layer.
- **Drafts count immediately.** Unpublished drafts count toward every repeat/partner rule the moment they exist, so two concurrent runs can't both claim the same pair.
- **One snapshot, many uses.** All diversity inputs (recent rosters, partner/opponent counts, overlap map) derive from a single per-slot history read — ordering enforced in SQL, fail-closed on error.
- **Checked-out players stay out.** Every restore path (end, cancel, clear, promote) guards against resurrecting a player who left back into the queue.

6 Audit findings — quirks the owner should know

All verified in code; none crash anything, but several are surprising:

1. **The Hard Cap guarantee excludes players with 5+ games.** A 30-minute waiter on their 5th game drops to Red Zone — players who've had their fair share have no absolute service guarantee (by design, protecting under-served players).
2. **Red Zone perks key off score, not wait.** Downstream checks use score ≥ 1000. A 20-minute waiter with heavy game debt can compute below 1000 and silently lose Red Zone treatment (wider skill windows, softer penalties).
3. **The cap-saturation banner can misattribute.** It fires whenever the partner-cap pre-filter removed anyone, even when the real blocker was skill-window exhaustion.
4. **One silent repeat path.** When a diversity violation's would-be swap target is themselves Red Zone, the repeat is accepted outright without being flagged — so the cross-court escape hatch never fires for it.
5. **A fully partner-capped pool produces no match at all**, even past the 15-minute fallback. Only manual organizer assignment gets around it (the “no waivers” rule working as intended).
6. **The rest-filter fallback is all-or-nothing.** If fewer than 4 rested players remain, the entire unrested pool becomes eligible again — not just enough bodies to reach 4.
7. **Held drafts wake up lazily.** A held draft only becomes promotable when a later match-end or cancel runs the readiness recompute — the engine itself never re-checks.
8. **Opposite failure postures, deliberately.** A failed pool read degrades softly (engine sees “nobody waiting”); a failed history read fails closed (run stops). Both correct for their blast radius.

7 Freshness vs. the partner cap — validated on real 18-player Thursdays

Question: does the snake draft produce fresh pairings *before* rule 7 (partner cap of 2), or does it wait for the cap to bind? **Answer: fresh first.** The cap is a backstop, never a trigger. Three freshness layers fire before it:

1. **Candidate scoring.** Anyone who recently partnered *or* faced the anchor takes **+10,000 per overlap unit**. Teammate and opponent both weight 2, so one recent game together = +20,000, which buries that candidate — long before the cap matters.
2. **Diversity rejection.** A four where 3+ shared a recent match is thrown out.
3. **Snake draft's 4-pass search.** Passes 1a/1b require **both** team pairs at count = 0 (never partnered this session). Only passes 2a/2b accept count < cap. The order is genuinely never-partnered → partnered-once → refuse at twice.

Scope limit: the snake draft only chooses among 3 possible splits of an already-chosen four. It cannot fetch different players — that happened upstream in candidate scoring. Most of the freshness you actually feel comes from layer 1, not the draft itself.

The one exception: splits are partitioned balanced-vs-lopsided first, and the freshness passes run over the balanced pool only. A within-cap repeat on balanced teams beats a fresh-but-lopsided split — the only case where the engine takes a 2nd-time pairing while a fresh option existed.

Measured on four 18-player Thursday sessions:

Session	Matches	Engine-made	Partnerships	Partner repeats	By engine
07/30	21	8	42	1 (match 18)	1
07/09	22	3	44	0	0
06/25	28	5	56	1 (match 26)	0 — *manual*
05/07	21	6	42	0	0

Across all four sessions the engine produced **exactly one** repeat partnership, and **zero** pairs ever reached the cap. At 18 players there are C(18,2) = 153 possible pairs and a session consumes only ~42–56, so the fresh supply never runs dry — **rule 7 effectively never binds at your session size.**

Two things this surfaced:

- **Opponent repeats are a different story:** 23–35 per session, some pairs facing each other **4 times**. The opponent cap of 2 is a soft preference and manual matches ignore it entirely. If anyone says "I keep playing the same people," it's almost certainly across the net, not on their own side.
- **Most Thursday matches are manual, not engine-made** — 13/21, 19/22, 23/28, 15/21. Every guarantee in this document applies only to the *auto* minority; manually created matches bypass the engine's rules completely, including the partner cap.

8 File map

Layer	File	Role
Pure algorithm	src/lib/matchmaking-core.ts	Scoring, group assembly, team split, diversity, draft cap
Data layer	src/lib/matchmaking-db.ts	Pool fetch, history snapshot + pure derivations, match-commit RPCs
Orchestration	src/app/actions/matchmaking.ts	Engine loop, gates, cross-court trigger, promotion, Call Next
Triggers	queue.ts · match-lifecycle.ts · match-drafts.ts · sessions.ts	17+ call sites re-running the engine after state changes